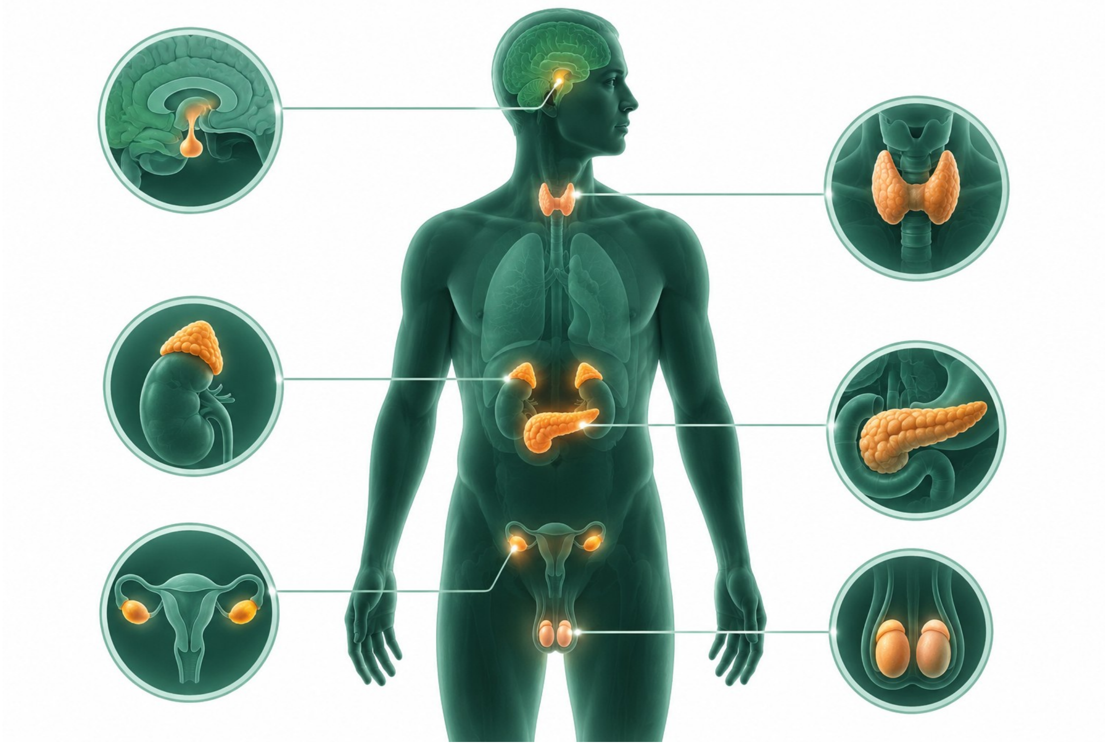
ENDOCRINE SYSTEM

NCLEX STUDY GUIDE

Hormones • Laboratory patterns • Emergencies • Diabetes

1. Endocrine Glands and Pituitary Function



Major endocrine glands: pituitary, thyroid, adrenals, pancreas and gonads

Anterior pituitary	Posterior pituitary
GH: dwarfism/acromegaly; TSH: thyroid; ACTH: adrenal cortex; FSH/LH: reproduction.	Oxytocin and ADH/vasopressin: fluid balance through DI and SIADH.

- Sheehan syndrome follows severe postpartum hemorrhage/hypotension: inability to lactate, amenorrhea, loss of pubic hair, fatigue, weight gain and hypotension; lifelong hormone replacement.
- Acromegaly: excess GH after maturity, enlarged hands/feet/jaw, coarse features, hyperglycemia, organ enlargement and visual-field changes. Treat pituitary tumor with surgery, octreotide or bromocriptine as ordered.
- After transsphenoidal hypophysectomy: HOB 30°, head midline, mouth breathing, avoid ICP-increasing activity; monitor CSF leak, DI and meningitis.

2. Diabetes Insipidus vs. SIADH

Diabetes insipidus	SIADH
Low ADH: polyuria, intense thirst, dehydration, hypotension. Urine SG <1.005; high sodium/osmolality/BUN/Hct. Treat fluids and desmopressin; goal is lower urine output and higher SG.	High ADH: low urine output, fluid overload, hypertension, confusion/seizure. Urine SG >1.030; low sodium/osmolality/BUN/Hct. Restrict fluid, diuretic, hypertonic saline for severe symptoms and demeclocycline as ordered.

NEURO PRIORITY Severe sodium change with confusion, seizure or cerebral edema is an emergency. Correct sodium carefully and monitor neurologic status.

3. Thyroid Disorders

Hypothyroidism	Hyperthyroidism / Graves
Cold intolerance, weight gain, dry skin, constipation, bradycardia, fatigue, slowed cognition and hyperlipidemia. T3/T4 low, TSH high. Levothyroxine on empty stomach in morning.	Heat intolerance, weight loss, diarrhea, diaphoresis, tachycardia, hypertension, tremor, insomnia and exophthalmos. T3/T4 high, TSH low. Methimazole/PTU, beta blocker and radioactive iodine as ordered.

Emergency	Key findings / response
Myxedema coma	Severe hypothermia, hypotension, hypoglycemia, hyponatremia, respiratory failure and coma. Airway, IV thyroid hormone, glucose, steroids and cautious warming.
Thyroid storm	High fever, marked tachycardia/AF, agitation, tremor, N/V and delirium. ICU, cooling, beta blocker, PTU/methimazole and iodine after antithyroid drug.

- Antithyroid drugs can cause agranulocytosis and liver injury; report fever or sore throat.
- Radioactive iodine is contraindicated in pregnancy; body fluids are radioactive for several days and require prescribed precautions.
- After thyroidectomy: semi-Fowler, support neck, monitor posterior bleeding, hypocalcemia/tetany and stridor; keep calcium gluconate and emergency airway equipment available.

4. Adrenal Disorders

Pheochromocytoma	Conn syndrome
Adrenal tumor with episodic/sustained severe hypertension, pounding headache, tachycardia, sweating and hyperglycemia. Do not palpate. VMA urine testing; alpha blockade/vasodilator and adrenalectomy.	Hyperaldosteronism: high sodium/fluid retention and low potassium with U waves/dysrhythmia. Give potassium and potassium-sparing diuretic; avoid agents that worsen hypokalemia.

Addison disease	Cushing syndrome
Low cortisol/aldosterone: hyperkalemia, hyponatremia, hypoglycemia, hypotension, weight loss, weakness and hyperpigmentation. Lifelong glucocorticoid/mineralocorticoid; increase steroid for stress.	Excess cortisol: moon face, buffalo hump, central weight gain, thin limbs, bruising/striae, infection, osteoporosis, hypertension, hyperglycemia and low K/Ca. High-protein, low-sodium/carbohydrate/calorie diet and safety.

ADDISONIAN CRISIS Hypotension, tachycardia, epigastric pain, confusion, severe hyponatremia and hypoglycemia: rapid IV isotonic fluids, IV hydrocortisone, glucose/electrolyte correction and monitoring.

5. Diabetes Mellitus and Metabolic Syndrome

Type 1	Type 2
Absolute insulin deficiency, often younger onset; requires insulin and is prone to DKA.	Insulin resistance/relative deficiency, often associated with central obesity; prone to HHS.

- Diagnostic patterns: fasting glucose ≥126 mg/dL or A1C ≥6.5% supports diabetes; A1C <7% is a common individualized control target.
- Classic findings: polyuria, polydipsia, polyphagia, weight loss, hyperglycemia, slow healing, infection and blurred vision.
- Long-term complications: retinopathy, nephropathy, neuropathy/foot injury and reproductive dysfunction.

6. Hypoglycemia and Hyperglycemic Emergencies

- Hypoglycemia (<70): hunger, sweating, tremor, palpitations, confusion, slurred speech, seizure/coma. If awake give 15-20 g rapid carbohydrate, recheck in 15 minutes and repeat; follow with carbohydrate/protein snack. If unable to swallow, use IV dextrose or glucagon as ordered.

DKA	HHS
Type 1; sudden, ketones, metabolic acidosis, Kussmaul breathing, fruity breath, abdominal pain and glucose often 300-600.	Type 2; gradual, profound dehydration and neurologic change, glucose >600/osmolality >350, usually no ketones.

TREATMENT ORDER Cardiac monitoring, IV isotonic fluid first, regular insulin IV, then add dextrose as glucose falls. Replace potassium when indicated and urine output is adequate.

7. Oral Diabetes Medications

Class	Safety point
Metformin	GI effects/lactic acidosis; check renal function and hold around iodinated contrast according to policy.
Sulfonylureas	Stimulate insulin; hypoglycemia and weight gain; caution with sulfa allergy/alcohol.
Thiazolidinediones	Fluid retention/heart failure risk.
Meglitinides	Short meal-time secretagogues; hypoglycemia.
DPP-4 inhibitors	Possible pancreatitis and liver concerns.
Alpha-glucosidase inhibitors	Give with first bite; GI effects; avoid with IBD/obstruction.

8. Insulin Administration

Type	Timing / mixing
Rapid: lispro/aspart/glulisine	Give with meal available; onset about 15 min, peak 1-3 hr.
Regular	Only insulin given IV; onset ~30-60 min, peak 2-4 hr; clear and may mix.
NPH	Cloudy; onset ~1-2 hr, peak 4-12 hr; may mix with regular.
Long acting: glargine	No pronounced peak; do not mix in same syringe.

- Rotate sites systematically; abdomen provides consistent absorption. Do not massage or apply heat/exercise to injection area.
- Store unopened insulin refrigerated; avoid freezing/direct sunlight; administer at room temperature and discard opened vial per manufacturer.
- Mixing regular/NPH: inject air into NPH, air into regular, withdraw clear regular first, then cloudy NPH.
- Dawn phenomenon: morning hyperglycemia without overnight low; may need more insulin. Somogyi: rebound after 2-3 AM hypoglycemia; bedtime snack or reduced evening NPH.
- Foot care: inspect daily, lukewarm wash/dry between toes, moisturizer not between toes, straight nail trimming, proper closed shoes, no barefoot walking or heating pads.

FINAL NCLEX FOCUS Prioritize airway/LOC changes, severe sodium or potassium abnormalities, thyroid/Addisonian crisis, hypoglycemia, DKA and HHS. Trend glucose, electrolytes, urine output and cardiac rhythm.